Several materials can improve the compatibility between natural rubber (NR) and EPDM, each with distinct mechanisms, typical examples, and cost profiles.

□ Modified EPDM (Maleic Anhydride Grafted)

This is the most widely used approach. The EPDM backbone is chemically modified with maleic anhydride (MAH) groups, which can react with polar groups or form covalent bonds at the interface .

* **Typical Examples**: **Exxelor VA 1801** (MAH-grafted EPDM, high grafting level) . **Coace** brand MAH-g-EPDM . CMG9802 (grafting degree 0.8%) from Sunrise Chemical .

* **Mechanism**: The grafted anhydride groups react with the NR phase or other polar additives, creating a chemical bridge that stabilizes the blend. Data shows modified EPDM significantly improves fatigue life, tear resistance, and tensile strength compared to unmodified EPDM .

* **Cost**: Prices vary by grade and supplier. For reference, MAH-g-EPDM products are listed around **$5.20/kg** (approx. ¥37/kg) on B2B platforms , with some specialized grades from Chinese distributors ranging from **¥16.2 to ¥48.8/kg** .

□ Silane Coupling Agents

These are primarily used to improve the bonding between fillers and the rubber matrix, which indirectly enhances blend compatibility.

* **Typical Example**: **STRUKTOL® SCA 1100** (gamma-Aminopropyltriethoxysilane) .

* **Mechanism**: The silane's ethoxy groups hydrolyze to form silanol groups that bond with hydroxyl-containing fillers (like silica or clay). The amino group then reacts with the rubber matrix. It is recommended for use with NR, EPDM, NBR, and their blends .

* **Cost**: Specific prices are not listed in the search results, but it is typically supplied in 40 lb pails or 440 lb drums .

□ MBS (Methyl Methacrylate-Butadiene-Styrene)

MBS is a core-shell impact modifier that can also act as a compatibilizer in certain blend systems.

* **Typical Examples**: **LG EM500** or **IM812** . Generic MBS powders from various Chinese suppliers .

* **Mechanism**: In NBR/EPDM blends, MBS provided the best swelling resistance and thermal stability compared to BIIR and zeolite, indicating its effectiveness as a compatibilizer .

* **Cost**: Market prices for MBS compatibilizers range from **¥22 to ¥32/kg**, depending on the grade and supplier .

□ Other Approaches

* **Bromobutyl Rubber (BIIR)**: Used as a compatibilizer for NBR/EPDM blends, it improved thermal and swelling characteristics .

* **Zeolite**: Also evaluated for NBR/EPDM compatibility, though MBS performed better in the cited study .

* **Hexamethylenediamine (HMD)**: Used as a curative for modified EPDM in NR blends to enhance co-vulcanization .

If you can share more details about your specific NR/EPDM formulation or the property you are most focused on, I can help narrow down the most suitable compatibilizer.